

SERGEY SHUVAEV

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Building AI with neural and cognitive priors to model human actions, preferences, and their neural basis
PhD in applied math and physics with a focus on computer science and computational neuroscience

Experience

Postdoctoral Researcher 3/25-now, *Dyer & Pesaran Labs*, University of Pennsylvania, Philadelphia, PA

- Develop neuro-AI models of neural and behavioral dynamics ([NeurIPS' 25](#)) for BMI
- Co-lead the development of interpretable time-series forecasting models ([arXiv '25](#))
- Apply these models to identify biomarkers and therapeutic targets in patient groups

Postdoctoral Researcher 11/22-2/25, **Student in Residence** 7/16-10/22,

Koulakov Lab, Cold Spring Harbor Laboratory, Cold Spring Harbor, NY

- Developed data-driven models of decision-making for stay-or-leave decisions (RL) ([NeurIPS '20](#)), motivation (RL) ([Front Sys Neurosci '21](#)), and conflict (Bayesian/game-theoretic) ([NeurIPS '23](#))
- Worked on a deep learning framework to predict the smell of odorants ([ICML '19](#)) and analyzed olfactory connectivity data to investigate how smell is processed in the brain ([PLOS Comp Bio '24](#))
- Co-developed methods for neural network compression ([PNAS '24](#)) and unfolding ([PNAS '19](#))

Research Associate 7/16-12/18, **Research Assistant** 1/12-6/16,

Enikolopov Lab, Moscow Institute of Physics and Technology, Moscow, Russia

- Developed automatic procedures to analyze cell populations in whole-brain samples: microscopy ([MethodsX '19](#)), 3D data alignment ([Sci Reports '22](#)), and object detection ([Front Neuroanat '17](#))
- Performed microscopy and analyzed data to evaluate common impacts on adult neurogenesis: irradiation ([NeuroReport '19](#)); antidepressants, brain development, and cell migration

Research Assistant 8/13-7/15, *Superconductivity Department*, Kurchatov Institute, Moscow, Russia

- Developed numerical models and worked towards experimental measurements of electro- and thermodynamics in high-current superconductive cables to pursue requirement-based design

Education

PhD ('22), **MSc** ('15), **BSc** ('13), **Physics and Mathematics**, Moscow Institute of Physics and Technology

Publications

- **Author:** 15+ papers (9 *first-auth.* incl. 2 [NeurIPS](#), 2 [PNAS](#); 2 *co-last*), 200+ citations; **see next page**
- **Reviewer:** [NeurIPS](#), [ICLR](#), [ICML](#), [COSYNE](#), [AAAI](#), [AISTATS](#); 5x+ *Top Reviewer*; **Guest Editor:** [PNAS](#)

Skills and qualifications

- Python, PyTorch, git, shell, W&B, LaTeX; prior work: Matlab, Wolfram Mathematica, C, C++
- Foundation models, time series modeling, sEEG, DBS, Bayesian inference, POMDP, reinforcement learning, game theory, computer vision, computational neuroscience, Theory-of-Mind

Awards

- PennAI Fellow, 2026. *Awarded to postdoctoral researchers with expertise in AI to bridge disciplines*
 - Top Reviewer: [ICLR '22](#); [NeurIPS '22](#), '24, '25, [AISTATS '24](#). *Awarded to top-5%/10% reviewers*
 - Travel awards: CSHL; Gatsby Charitable, Burroughs Wellcome, Google DeepMind, Simons, 2020
 - Swartz Fellow in Computational Neuroscience, 2016-2017. *\$100k+/2yrs toward salary and travel*
 - Alexandrov Scholar, 2012-2015. *Awarded to students with recent conference records & top-tier GPA*
 - Abramov and Frolov Scholar, 2010-2012. *Awarded to undergraduate students with the 4.0 GPA*
 - Kurchatov Award for Outstanding Research, 2013
- July 2026

Selected publications

PRISM: A Hierarchical Multiscale Approach for Time Series Forecasting ([arXiv '25](#))

*Chen, Z., Andre, A., Ma, W., Knight, I., **Shuvaev, S.**, and Dyer, E.*

A scalable self-supervised method for modeling human intracranial recordings during natural behavior ([NeurIPS '25 spotlight @BrainBodyFM](#))

*Mahato, S., Xiao, J., Andre, A., Chau, G., Ma, W., Knight, I., Nguyen, D., Hu, L., Brunton, B., Beauchamp, M., Pesaran, B., **Shuvaev, S.**, and Dyer, E.*

Encoding innate ability through a genomic bottleneck ([PNAS '24](#))

***Shuvaev, S.**, Lachi, D., Koulakov, A., and Zador, A.*

The primacy model and the structure of olfactory space ([PLOS Comp Bio '24](#))

*Giaffar, H., **Shuvaev, S.**, Rinberg, D., and Koulakov, A.*

A normative theory of social conflict ([NeurIPS '23](#))

***Shuvaev, S.**, Amelchenko, E., Smagin, D., Kudryavtseva, N., Enikolopov, G., and Koulakov, A.*

Spatiotemporal 3D image registration for mesoscale studies of brain development ([Sci Reports '22](#))

***Shuvaev, S.**, Lazutkin, A., Kiryanov, R., Anokhin, K., Enikolopov, G., and Koulakov, A.*

Neural networks with motivation ([Front Sys Neurosci '21](#))

***Shuvaev, S.**, Tran, N., Stephenson-Jones, M., Li, B., and Koulakov, A.*

R-learning in actor-critic model offers a biologically relevant mechanism for sequential decision-making ([NeurIPS '20](#))

Shuvaev, S., Starosta, S.*, Kvitsiani, D., Kepecs, A., and Koulakov, A.*

DeepNose: Using artificial neural networks to represent the space of odorants ([ICML '19](#))

*Tran, N., Kepple, D., **Shuvaev, S.**, and Koulakov, A.*

Network cloning using DNA barcodes ([PNAS '19](#))

***Shuvaev, S.**, Başerdem, B., Zador, A., and Koulakov, A.*

Click histochemistry for whole-mount staining of brain structures ([MethodsX '19](#))

*Lazutkin, A., **Shuvaev, S.**, and Barykina, N.*

Suppressed neurogenesis without cognitive deficits: effects of fast neutron irradiation in mice ([NeuroReport '19](#))

*Mineyeva, O., Barykina, N., Bezriadnov, D., ..., **Shuvaev, S.**, Usova, S., and Lazutkin, A.*

DALMATIAN: an algorithm for automatic cell detection and counting in 3D ([Front Neuroanat '17](#))

***Shuvaev, S.**, Lazutkin, A., Kedrov, A., Anokhin, K., Enikolopov, G., and Koulakov, A.*

Representations of sound in deep learning of audio features from music ([arXiv '17](#))

***Shuvaev, S.**, Giaffar, H., and Koulakov, A.*

Details: scholar.google.com/citations?user=2u5090wAAAAJ